BOOBALAN KANNAN

kboobalan2005@mail.com | +91 6379827940 | GitHub | LinkedIn

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience in Generative AI, RAG pipelines, and MCP integrations using LangChain and FastAPI. Proficient in engineering computer vision models, fine-tuning deep learning architectures, and building asynchronous backends. Passionate about deploying production-grade AI solutions to solve complex problems.

TECHNICAL SKILLS

Programming Languages	Python, Java, SQL
AI & Machine Learning	Machine Learning, Deep Learning, Supervised/Unsupervised Learning, Computer Vision, GenAI, RAG, MCP
Libraries & Frameworks	NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras, LangChain, LangGraph
Data Analysis	Data Cleaning, Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA)
Databases & Backend	MySQL, PostgreSQL, FastAPI, Celery
Tools & Web	Git, GitHub, VS Code, Jupyter Notebook, Google Colab, HTML, CSS, Bootstrap

PROFESSIONAL EXPERIENCE

CODEWORK – AI Trainee Engineer MAR 2026 – AUG 2026

- Developed RAG-based LLM applications using LangChain and LangGraph for context-aware AI solutions.
- Built FastAPI-based AI backend services and integrated LLM workflows with application APIs.
- Debugged and optimized AI workflows to improve API reliability, response handling, and overall application performance.

PROJECTS

Interactive Bible Project – Backend Development | FastAPI, Celery, PostgreSQL **GitHub**

- Contributed to API development, database integration, and AI-powered features including video, quiz, report, image, and audio generation.
- Implemented asynchronous background processing with Celery and PostgreSQL to handle concurrent user requests without API timeouts.

Arabic Handwritten OCR | Python, TensorFlow, Keras, CNN, Computer Vision **GitHub**

- Developed a deep learning-based OCR system utilizing CNN and Transformer models to recognize Arabic handwritten text from scanned document images.
- Applied image preprocessing, data augmentation, segmentation, and normalization techniques to enhance text recognition accuracy.

Stroke Lesion Segmentation & Classification | Python, TensorFlow, Keras, U-Net, CNN **GitHub**

- Trained U-Net and CNN architectures to detect and segment stroke lesions from brain MRI scan images with high precision.
- Performed extensive data preprocessing and evaluated model efficiency using Dice Score, F1-Score, and ROC-AUC metrics.

Weather Application using MCP | Python, MCP, OpenWeather API, FastAPI **GitHub**

- Engineered a real-time weather application leveraging Model Context Protocol (MCP) and OpenWeather API to retrieve city-wide meteorological data.
- Integrated live API data streams to display detailed temperature, humidity, weather conditions, and wind speed parameters cleanly.

Oman AI Trip Planner Chatbot | Python, LLM, FastAPI, REST APIs

- Built an LLM-powered trip planning assistant for Oman that integrates travel, weather, hotel, and restaurant APIs for personalized itineraries.
- Documented API integration architecture and system data flow to ensure reliable operation and future application expansion.

EDUCATION

B.E. Computer Science and Engineering
Sri Shanmugha College of Engg. & Tech., Salem 2022 – 2026 | CGPA: 7.54

HSC
Govt. Boys Hr. Sec. School, Kumarapalayam 2021 – 2022 | 63%

SSLC
Reliance Metric Hr. Sec. School, Kumarapalayam 2019 – 2020 | 79%

CERTIFICATIONS

- **Neural Networks & Deep Learning** – NPTEL **[Certificate]**
- **Introduction to Machine Learning** – Infosys Springboard **[Certificate]**
- **AI/ML Training Program** – CPLC **[Certificate]**